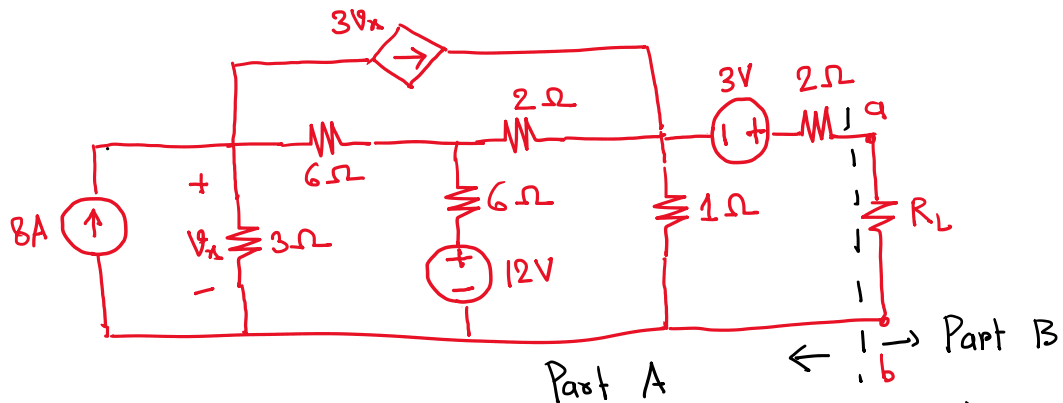
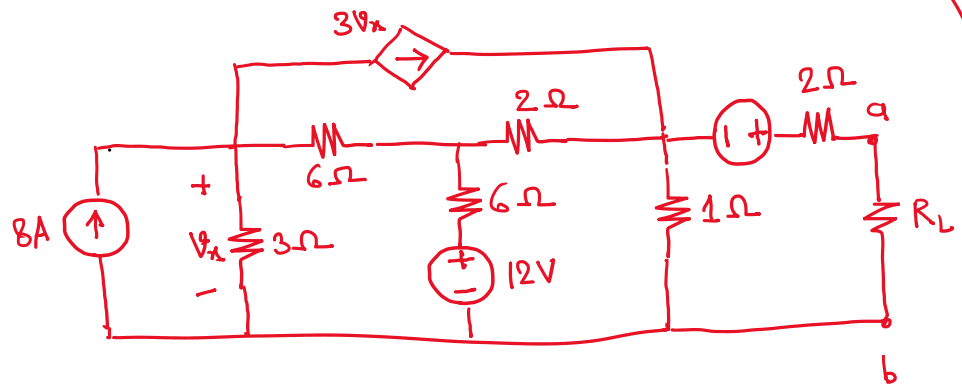


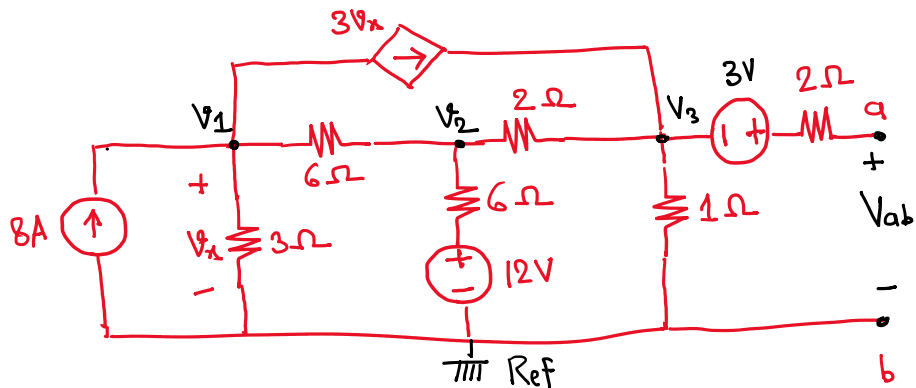
Thevenin's Soln1

Friday, December 19, 2025 7:05 PM

V_{TH} , R_{TH}



Remove Part B. (only Part A Remaining)



$$V_{TH} = V_{OC} = V_{ab}$$

$$\text{Node 1: } \frac{V_1}{3} + \frac{V_1 - V_2}{6} - 8 + 3V_x = 0 \quad V_x = V_1 - 0 = V_1$$

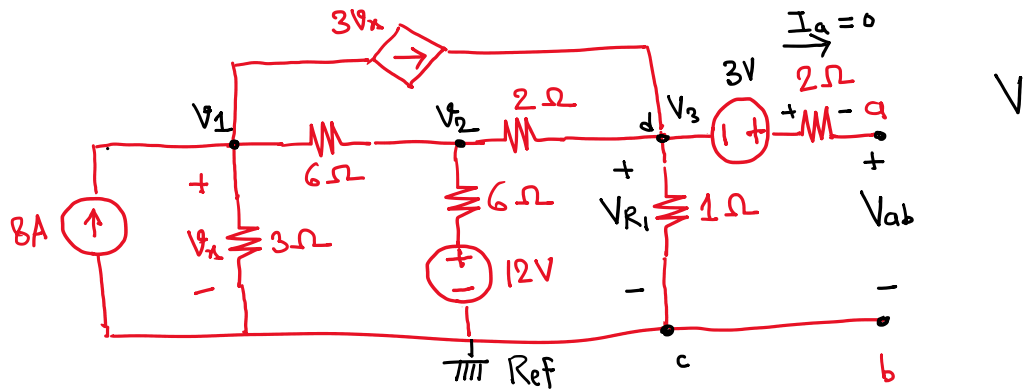
$$\Rightarrow \frac{V_1}{3} + \frac{V_1 - V_2}{6} - 8 + 3V_1 = 0 \quad \text{--- (I)}$$

$$\text{Node 2: } \frac{V_2 - V_1}{6} + \frac{V_2 - V_3}{2} + \frac{V_2 - 12 - 0}{6} = 0 \quad \text{--- (II)}$$

$$\text{Node 3: } \frac{V_3 - 0}{1} + \frac{V_3 - V_2}{2} - 3V_x = 0$$

$$\frac{V_3 - 0}{1} + \frac{V_3 - V_2}{2} - 3V_1 = 0 \quad \text{--- (iii)}$$

V_1, V_2, V_3 [After solving the equations]

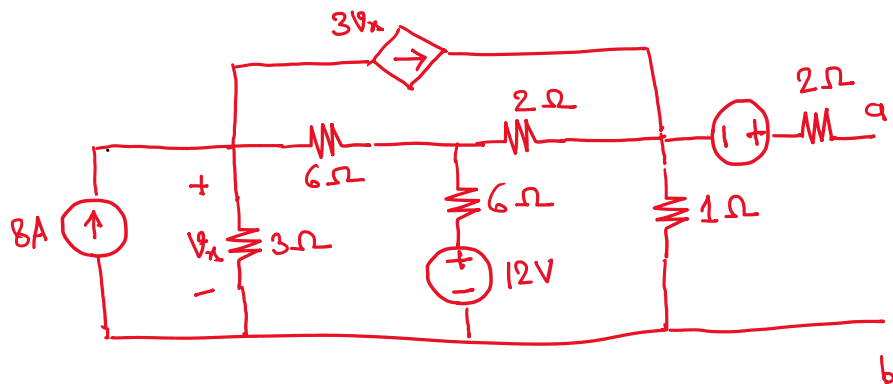


Applying KVL around loop $\overset{c}{\text{---}} \overset{d}{\text{---}} \overset{a}{\text{---}} \overset{b}{\text{---}}$

$$+V_{R1} + 3 - 2I_a - V_{ab} = 0 \quad \Bigg| \quad V_{R1} = V_3 - 0 = V_3$$

$$\therefore V_3 + 3 - V_{ab} = 0$$

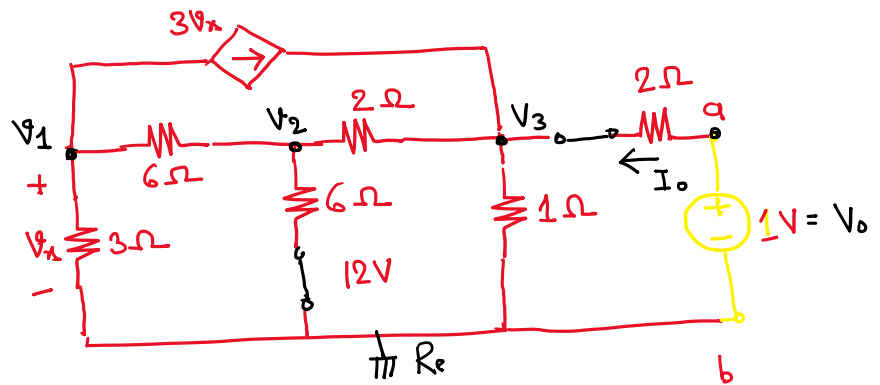
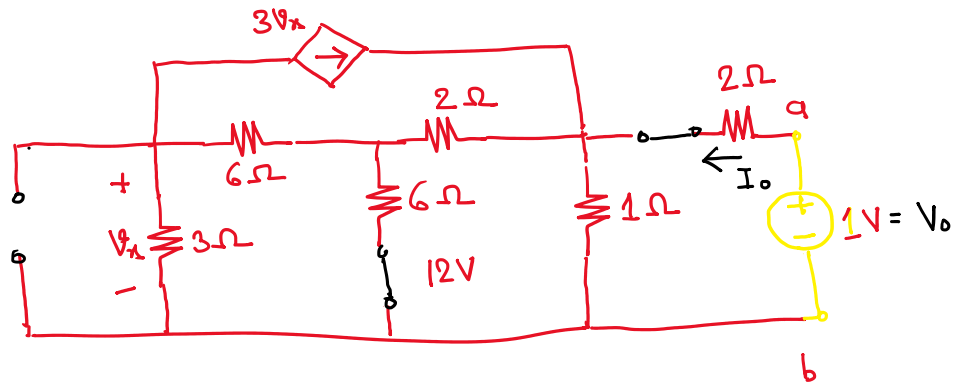
$$V_{Th} = V_{oc} = V_{ab} = V_3 + 3$$



To determine R_{Th} : 1. Remove Independent sources

2. IF dependent source exists, place an arbitrary Voltage source (say $V_0 = 1V$) between the nodes a & b





Apply Node Analysis method to determine V_3

$$I_0 = \frac{V_0 - V_3}{2} = \frac{1 - V_3}{2}$$

$$R_{TH} = \frac{V_0}{I_0}$$

